实验记录_20260127_目标伯恩斯坦椭圆区域宽度越窄，拟合效果会不会越差

🍶 **实验名称：**目标伯恩斯坦椭圆区域宽度越窄，拟合效果会不会越差

实验时间：20260127

colab代码：
https://colab.research.google.com/drive/1IRfOCpb6TksrCc5IsQ4brKbI6C4j4QA2?usp=sharing

实验结论：符合预期，目标伯恩斯坦椭圆区域越窄（beta越小），拟合效果越差

实验目的

验证当目标伯恩斯坦椭圆区域窄到没有包含奇点的时候，拟合效果会变差。

我们的猜想是学习到的伯恩斯坦椭圆和目标伯恩斯坦椭圆区域一致的时候，拟合效果最好，而当目标函数的伯恩斯坦椭圆区域太窄，窄到都无法包含奇点，那么神经网络的伯恩斯坦椭圆区域几乎没办法和目标伯恩斯坦椭圆区域一样，自然拟合效果会变差

如果结果符合预期，就进一步说明解析区域和优化有关

实验要点

1. 如何使得伯恩斯坦椭圆区域变窄

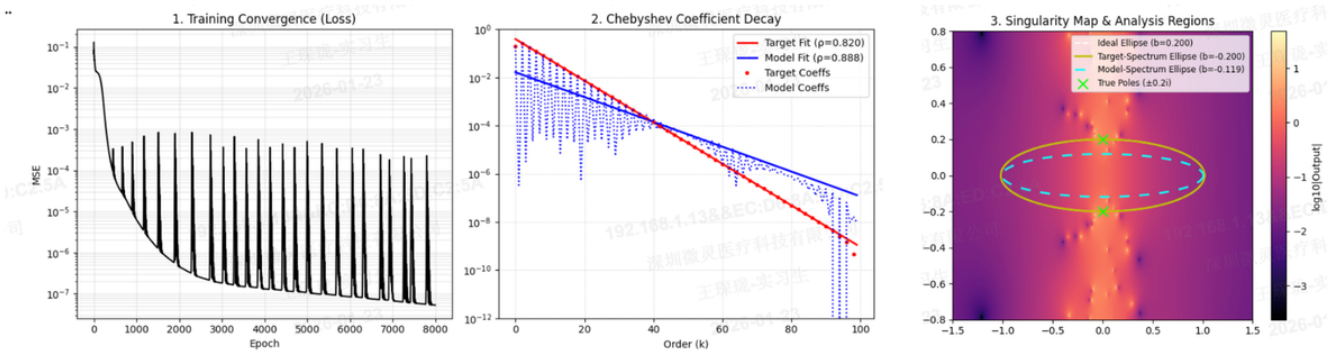
我们之前的目标函数是龙格函数 $\frac{1}{1+25x^2}$，这个时候奇点是0.2i（分母为0），可以很容易想到如果我们改变分母项，奇点也就会随之改变了。所以我们定义广义龙格函数 $f(x)=\frac{1}{1+(x/\beta)^2}=\frac{\beta^2}{\beta^2+x^2}$，这个函数的奇点为 βi。我们只要将 β 调小，伯恩斯坦椭圆区域就变窄了。

2. 拟合效果用什么表示

指标：mse loss

3. 可视化图

首先我会设置一组beta值（[0.5, 0.2, 0.1, 0.05]），以表示越来越窄的目标伯恩斯坦椭圆区域。仍然会输出每个函数下的loss图、切比雪夫系数图、奇点图和拟合图（用来训练的真实值和预测值，带上loss），像这样



最后还将所有loss绘制成beta vs loss的图

实验流程

龙格函数

2000个切比雪夫极值点 $x_k = \cos\left(\frac{k}{n}\pi\right)$

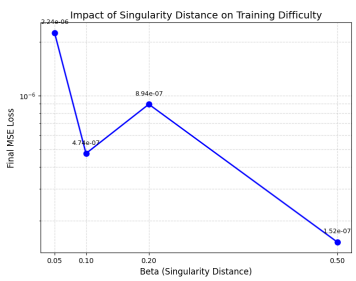
两层隐藏层的mlp（为了产生网络奇点在目标伯恩斯坦椭圆区域内部的情况）

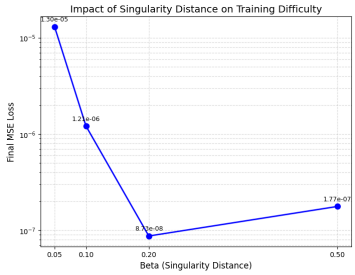
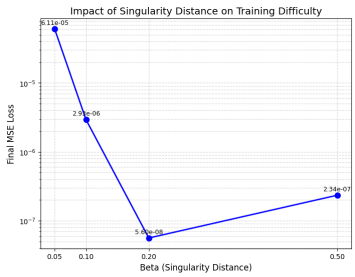
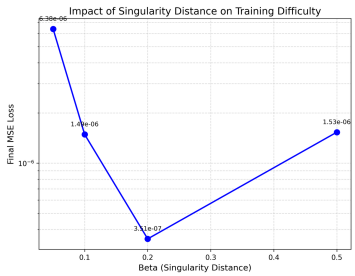
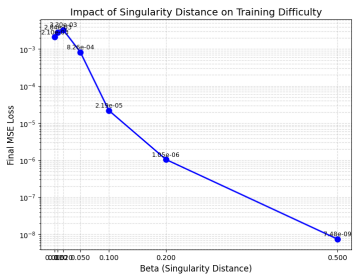
图片：

loss图，拟合图，频谱系数图，奇点图

最后再来一个汇总图：train loss vs beta图

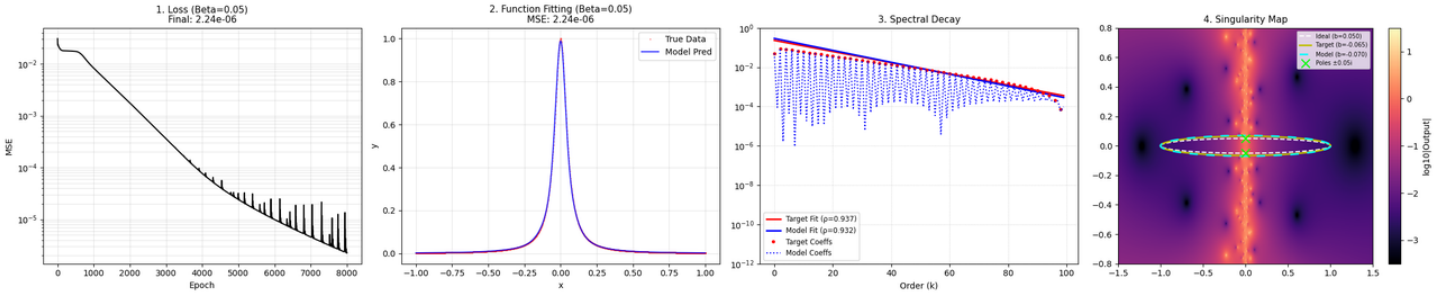
实验组

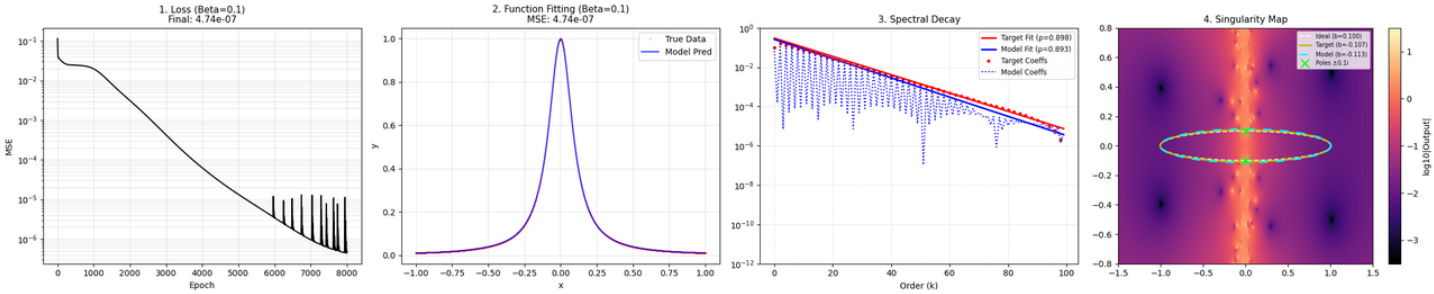
序号	实验目的	训练轮次	beta列表	神经元个数	学习率	实验结果	实验图
1	验证网络伯恩斯坦椭圆区域比目标伯恩斯坦椭圆区域宽时，拟合效果会变差	8000	[0.5, 0.2, 0.1, 0.05]	128	1e-4	最窄的伯恩斯坦椭圆区域确实对应最大的loss，**大致的趋势是区**	

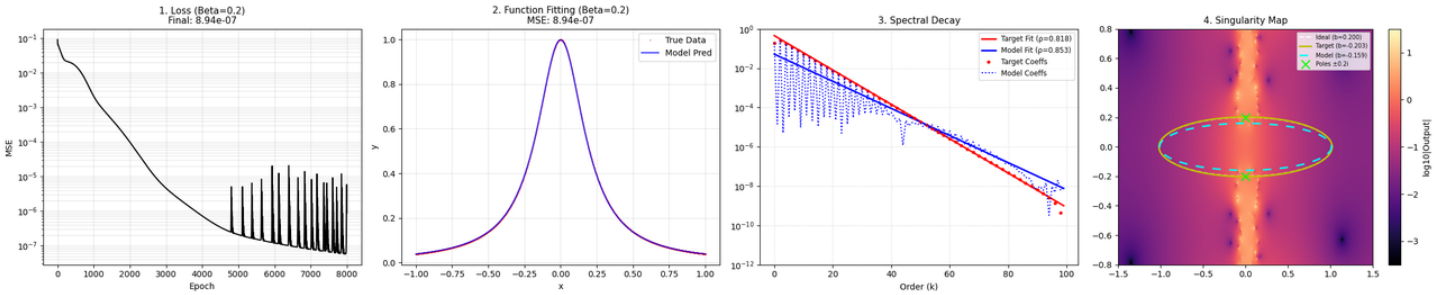
						域越宽，loss越小。	
2		10000			1e-4	**增大训练轮次后发现beta为0.2的目标函数对应的loss最小**	
3		10000			5e-5	同上	
4		6000					
5		20000	[0.5, 0.2, 0.1, 0.05,0.02,0.01,0.005]		1e-5	符合预期当伯恩斯坦椭圆区域变窄，拟合效果会变差 beta为0.02对应最差的效果，此时loss为3.3e-3 beta为0.5对应最好的效果，此时loss为7.48e-9	

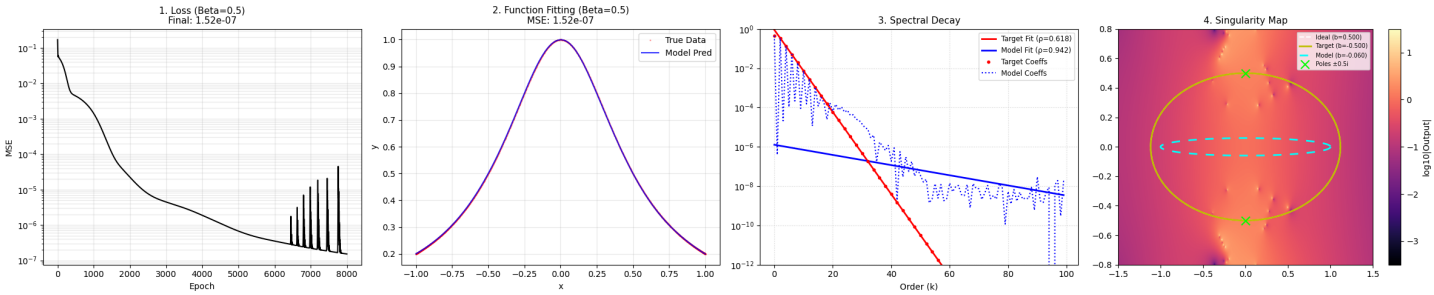
实验结果

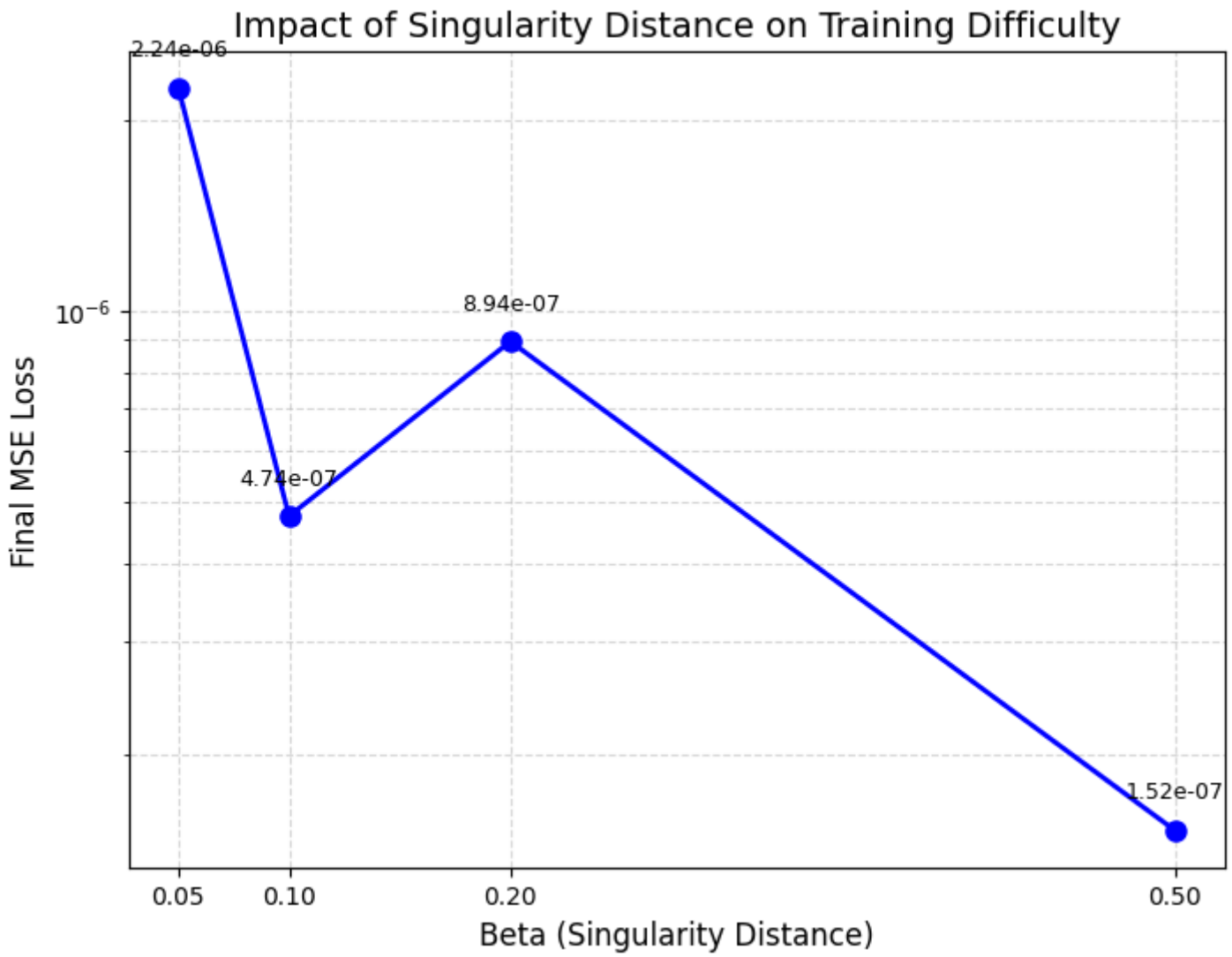
lr=0.0001, epoch8000



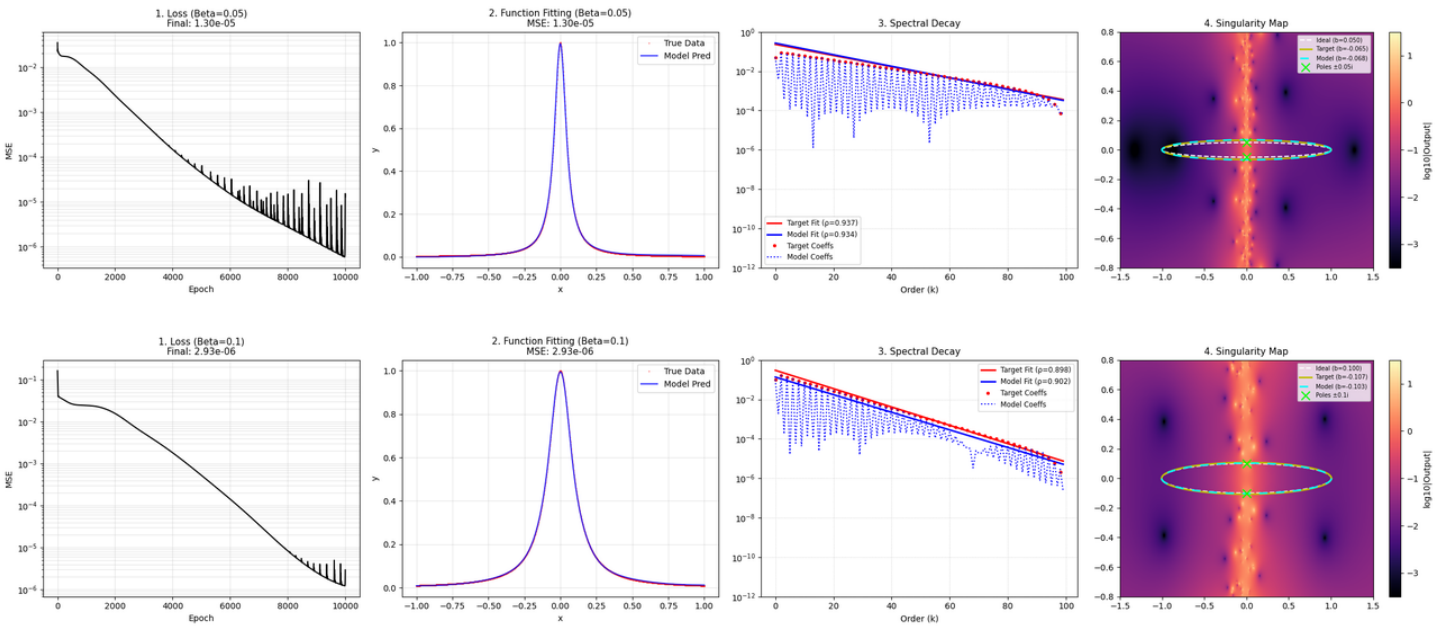


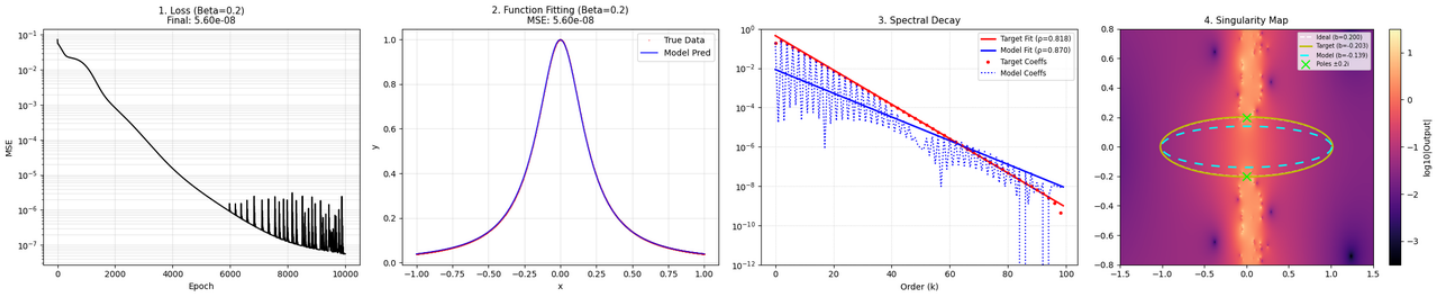


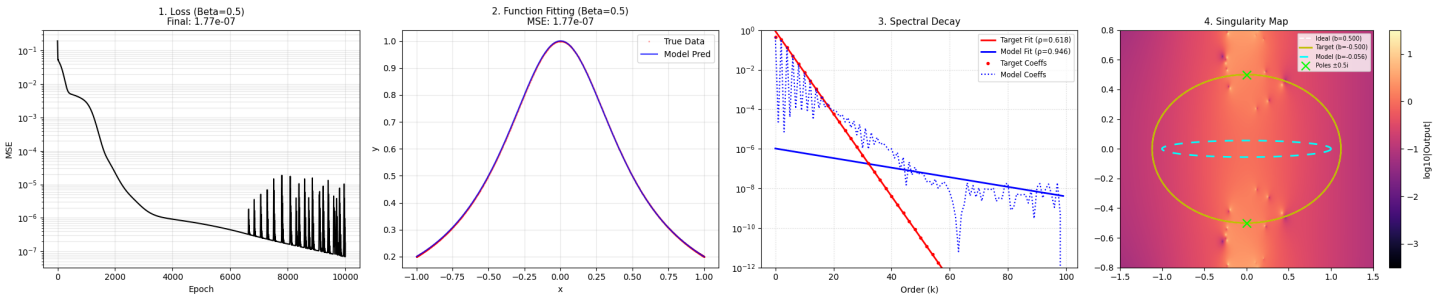


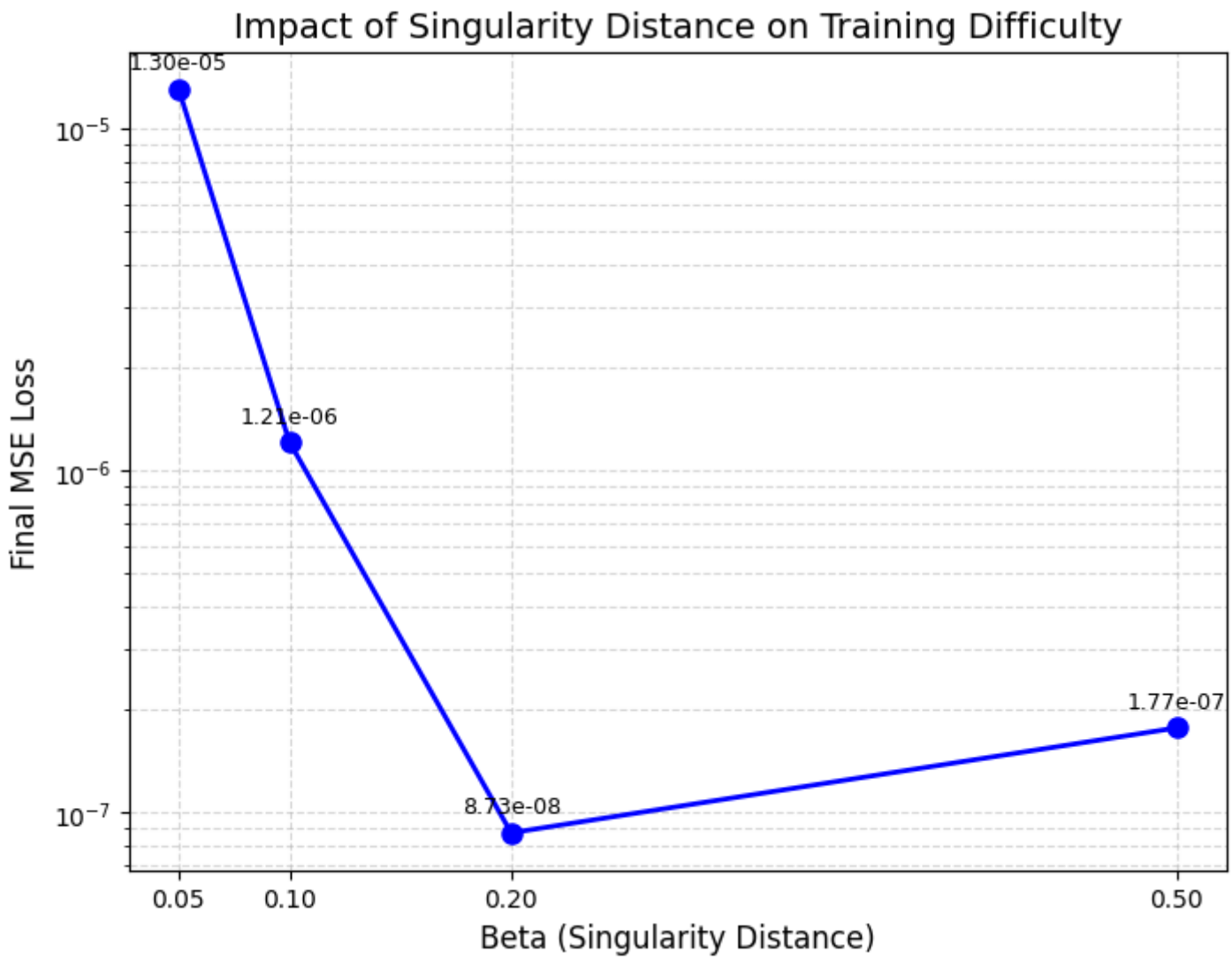


lr=0.0001，epoch=10000

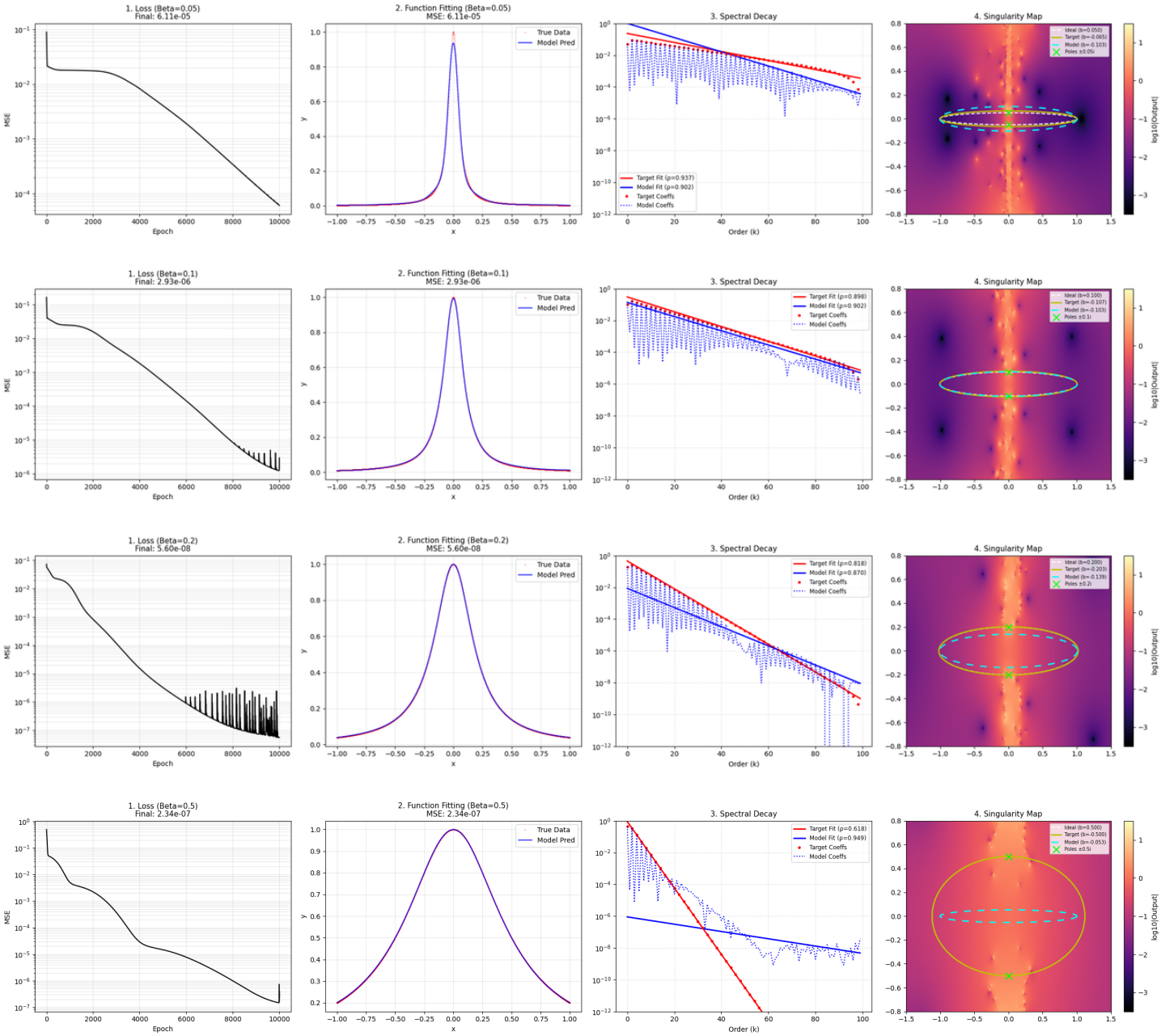


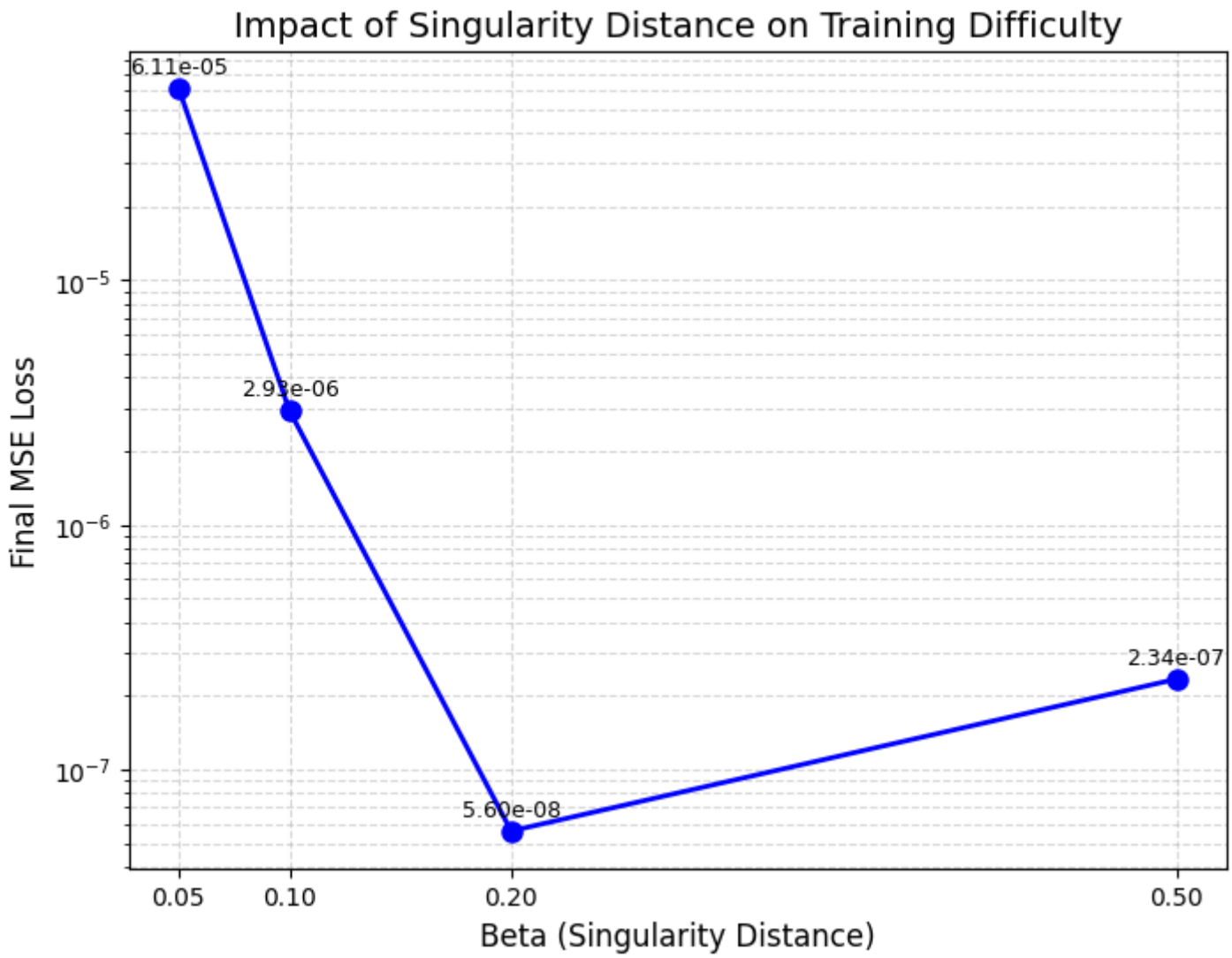




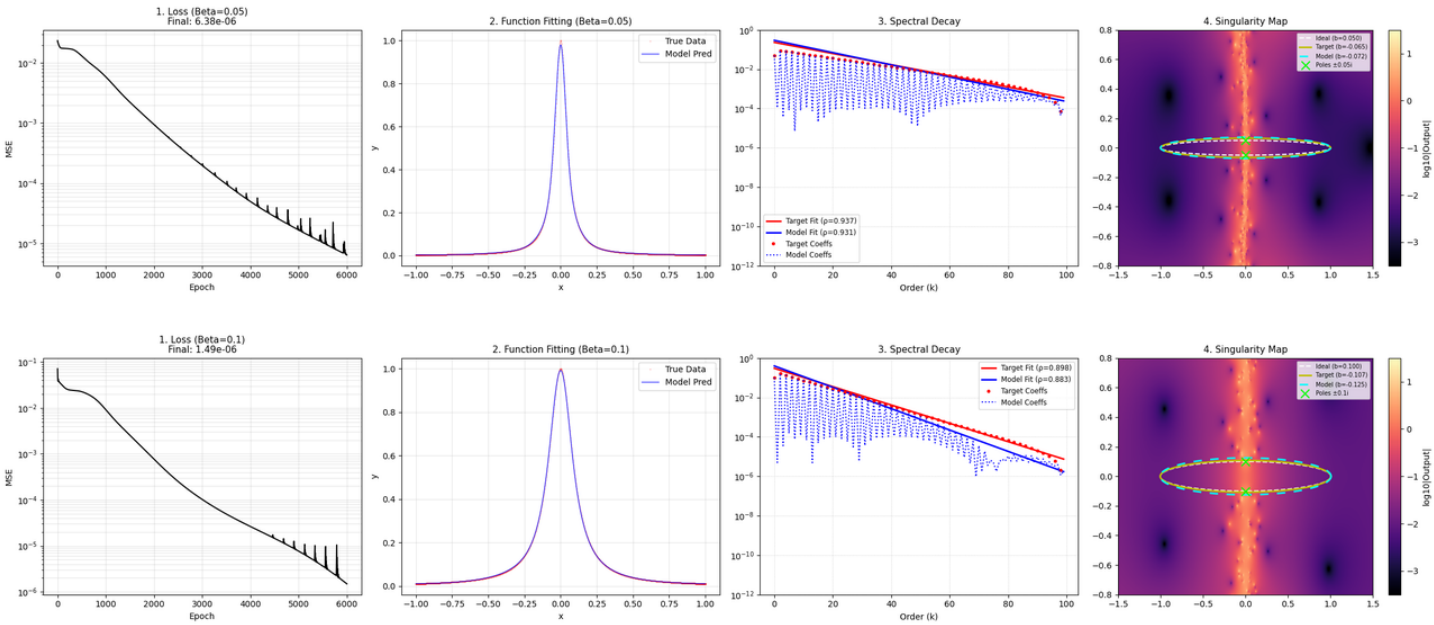


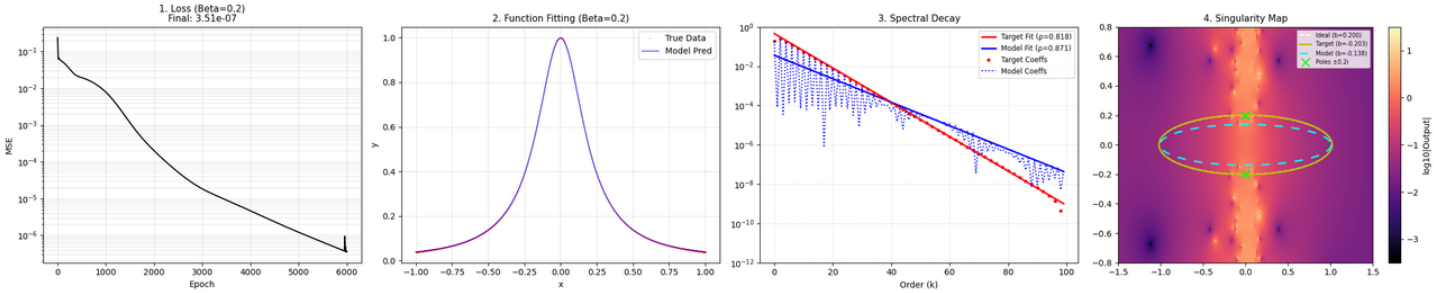
Lr=5e-5, epoch=10000

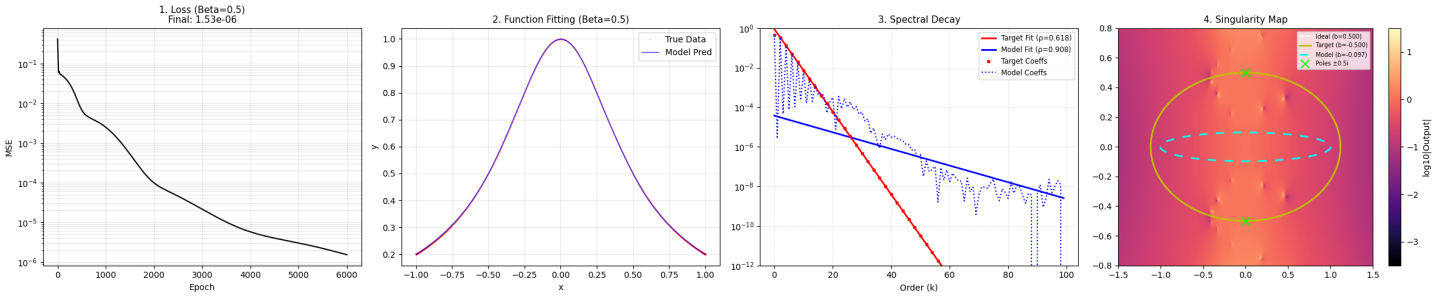

1. Loss (Beta=0.05)
Final: 6.11e-05
MSE
Epoch
2. Function Fitting (Beta=0.05)
MSE: 6.11e-05
True Data
Model Pred
x
y
3. Spectral Decay
Target Fit (ρ=0.937)
Model Fit (ρ=0.902)
Target Coeffs
Model Coeffs
Order (k)
4. Singularity Map
Ideal (b=0.050)
Target (b=-0.065)
Model (b=-0.103)
Poles ±0.05i
log10|Output|
1. Loss (Beta=0.1)
Final: 2.93e-06
MSE
Epoch
2. Function Fitting (Beta=0.1)
MSE: 2.93e-06
True Data
Model Pred
x
y
3. Spectral Decay
Target Fit (ρ=0.898)
Model Fit (ρ=0.902)
Target Coeffs
Model Coeffs
Order (k)
4. Singularity Map
Ideal (b=0.100)
Target (b=-0.107)
Model (b=-0.103)
Poles ±0.1i
log10|Output|
1. Loss (Beta=0.2)
Final: 5.60e-08
MSE
Epoch
2. Function Fitting (Beta=0.2)
MSE: 5.60e-08
True Data
Model Pred
x
y
3. Spectral Decay
Target Fit (ρ=0.818)
Model Fit (ρ=0.870)
Target Coeffs
Model Coeffs
Order (k)
4. Singularity Map
Ideal (b=0.200)
Target (b=-0.203)
Model (b=-0.139)
Poles ±0.2i
log10|Output|
1. Loss (Beta=0.5)
Final: 2.34e-07
MSE
Epoch
2. Function Fitting (Beta=0.5)
MSE: 2.34e-07
True Data
Model Pred
x
y
3. Spectral Decay
Target Fit (ρ=0.618)
Model Fit (ρ=0.949)
Target Coeffs
Model Coeffs
Order (k)
4. Singularity Map
Ideal (b=0.500)
Target (b=-0.500)
Model (b=-0.053)
Poles ±0.5i
log10|Output|

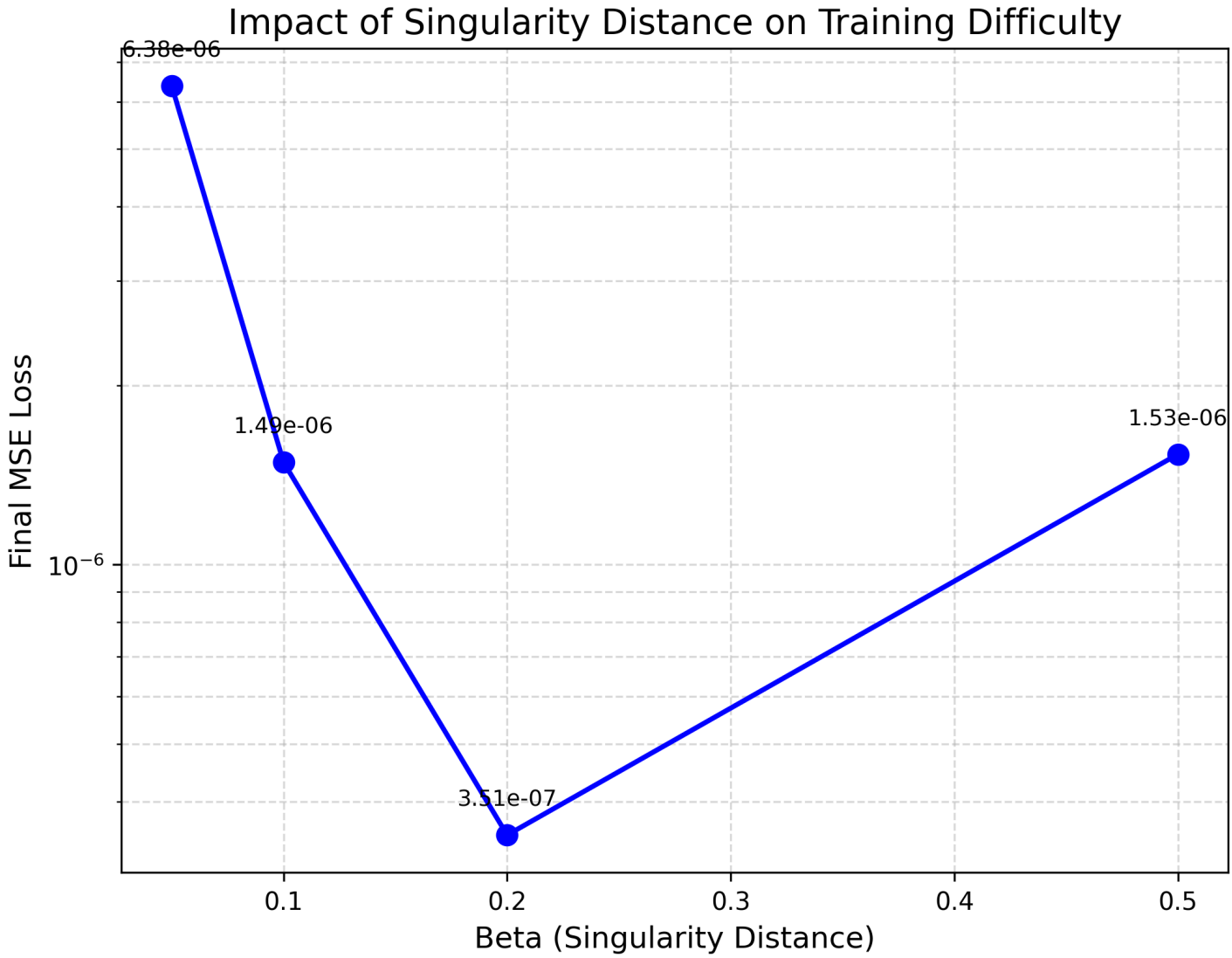


Lr = 1e-4 epochs=6000

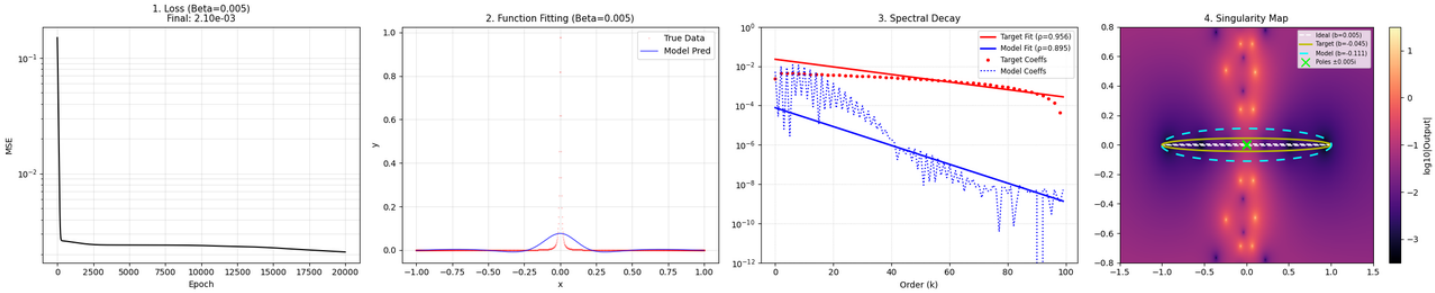


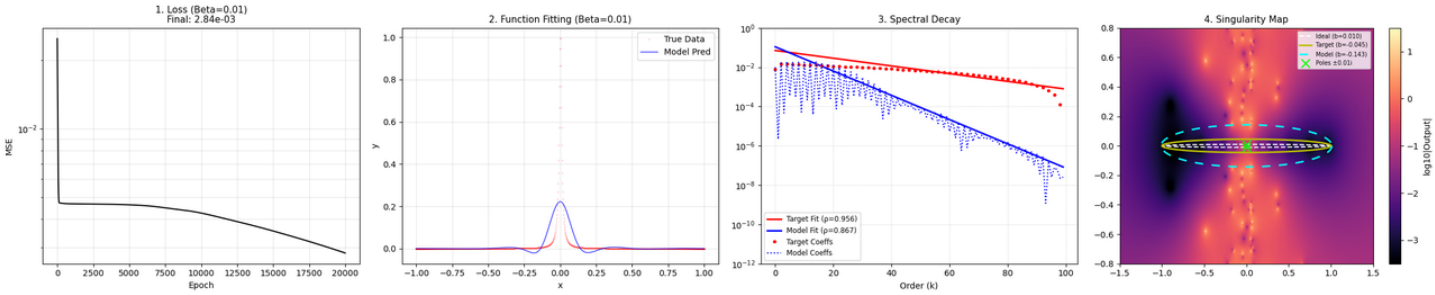


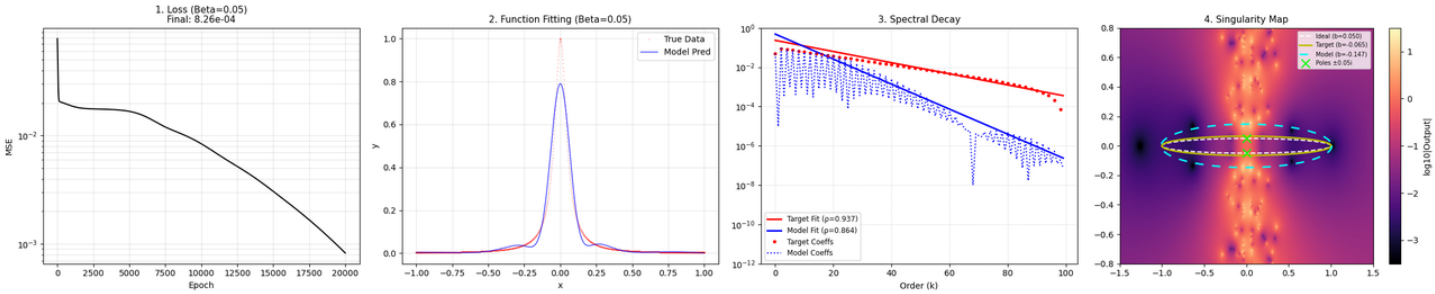


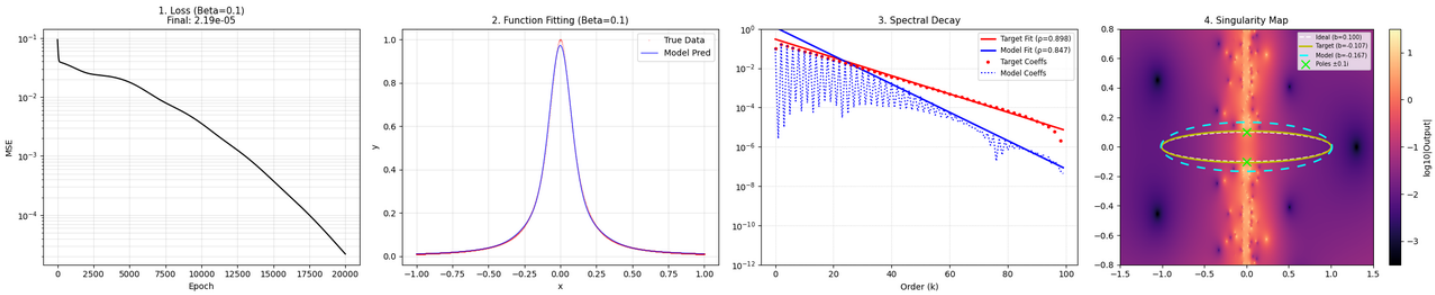


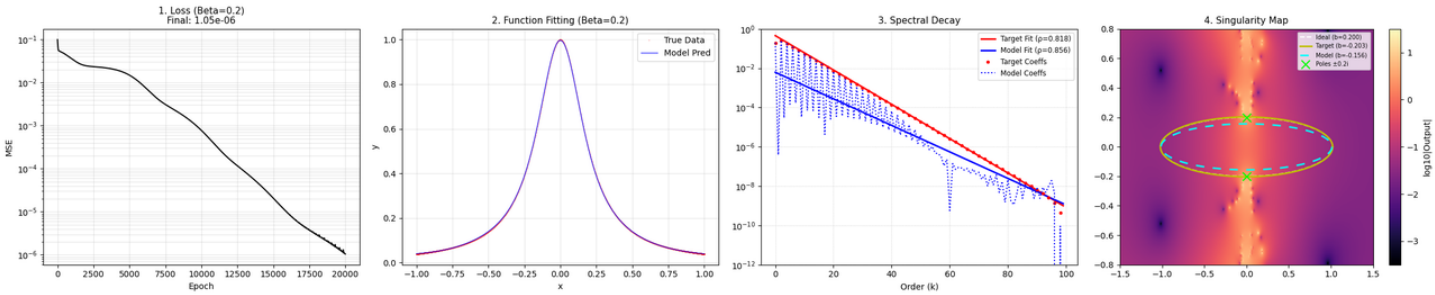
beta = [0.5, 0.2, 0.1, 0.05, 0.02, 0.01, 0.005] , lr = 1e-5, epochs = 20000

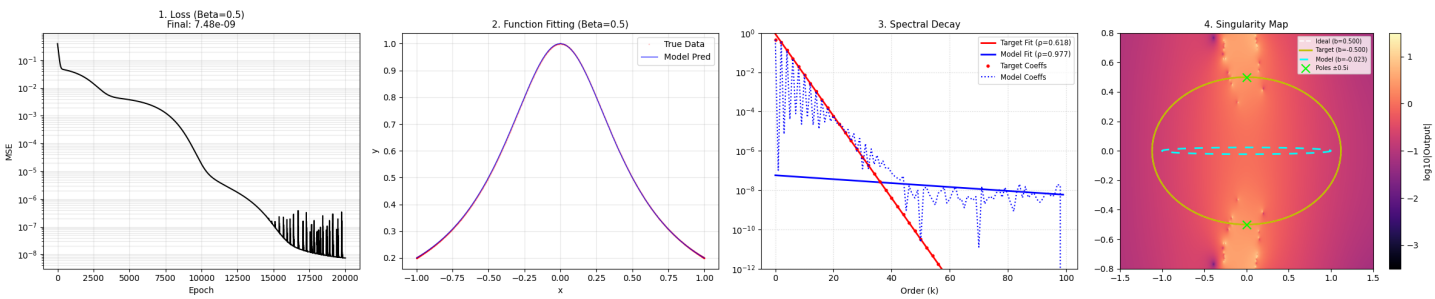

1. Loss (Beta=0.005)
Final: 2.10e-03
MSE
Epoch
2. Function Fitting (Beta=0.005)
True Data
Model Pred
y
x
3. Spectral Decay
Target Fit (ρ=0.956)
Model Fit (ρ=0.895)
Target Coeffs
Model Coeffs
Order (k)
4. Singularity Map
Ideal (b=0.005)
Target (b=-0.045)
Model (b=-0.111)
Poles ±0.005i
log10|Output|

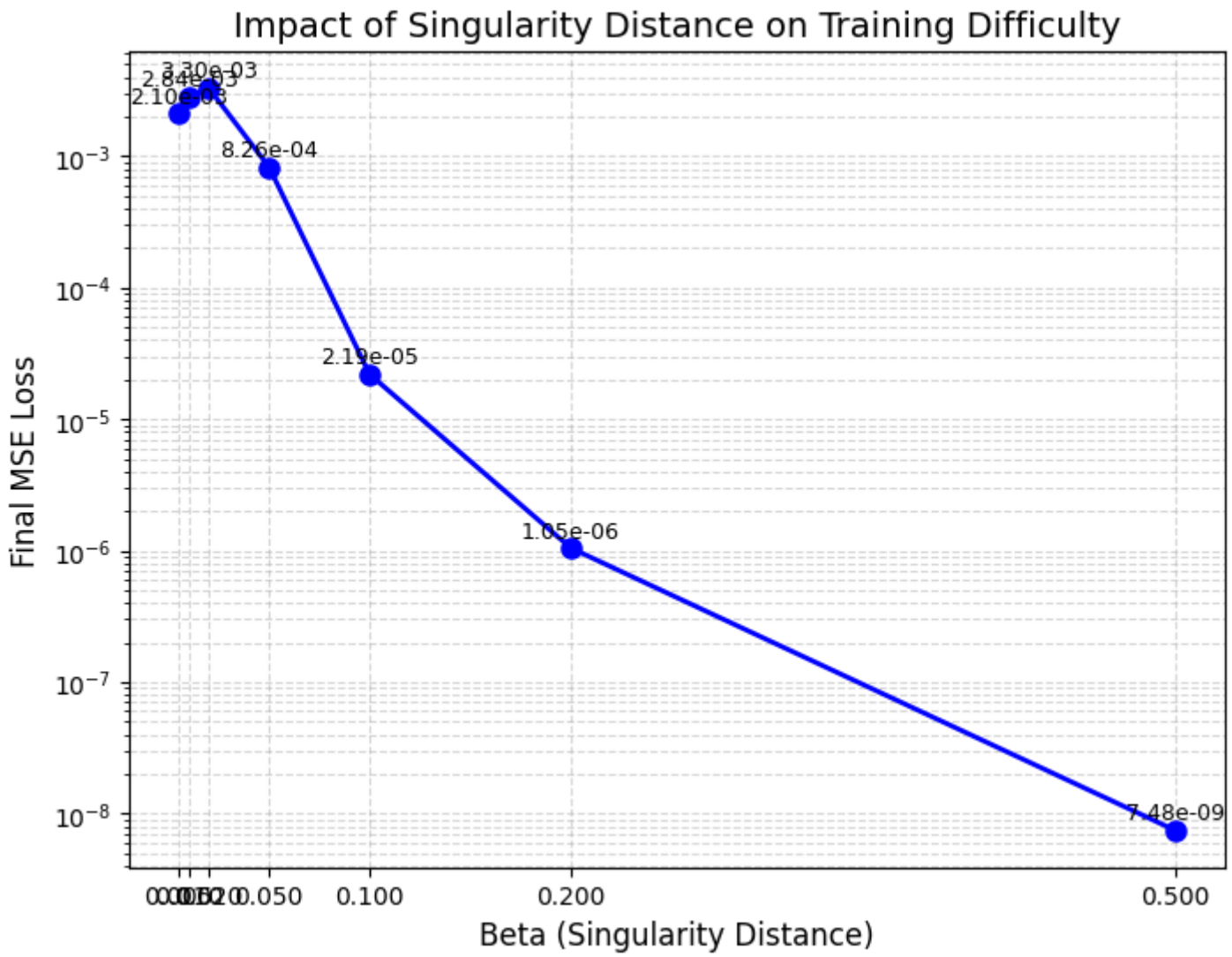

1. Loss (Beta=0.01)
Final: 2.84e-03
MSE
Epoch
2. Function Fitting (Beta=0.01)
True Data
Model Pred
y
x
3. Spectral Decay
Target Fit (ρ=0.956)
Model Fit (ρ=0.867)
Target Coeffs
Model Coeffs
Order (k)
4. Singularity Map
Ideal (b=0.010)
Target (b=-0.045)
Model (b=-0.143)
Poles ±0.01i
log10|Output|


1. Loss (Beta=0.05)
Final: 8.26e-04
MSE
Epoch
2. Function Fitting (Beta=0.05)
True Data
Model Pred
y
x
3. Spectral Decay
Target Fit (ρ=0.937)
Model Fit (ρ=0.864)
Target Coeffs
Model Coeffs
Order (k)
4. Singularity Map
Ideal (b=0.050)
Target (b=-0.065)
Model (b=-0.147)
Poles ±0.05i
log10|Output|


1. Loss (Beta=0.1)
Final: 2.19e-05
MSE
Epoch
2. Function Fitting (Beta=0.1)
True Data
Model Pred
y
x
3. Spectral Decay
Target Fit (ρ=0.898)
Model Fit (ρ=0.847)
Target Coeffs
Model Coeffs
Order (k)
4. Singularity Map
Ideal (b=0.100)
Target (b=-0.107)
Model (b=-0.167)
Poles ±0.1i
log10|Output|


1. Loss (Beta=0.2)
Final: 1.05e-06
MSE
Epoch
2. Function Fitting (Beta=0.2)
True Data
Model Pred
y
x
3. Spectral Decay
Target Fit (ρ=0.818)
Model Fit (ρ=0.856)
Target Coeffs
Model Coeffs
Order (k)
4. Singularity Map
Ideal (b=0.200)
Target (b=-0.203)
Model (b=-0.156)
Poles ±0.2i
log10|Output|


1. Loss (Beta=0.5)
Final: 7.48e-09
MSE
Epoch
2. Function Fitting (Beta=0.5)
True Data
Model Pred
y
x
3. Spectral Decay
Target Fit (ρ=0.618)
Model Fit (ρ=0.977)
Target Coeffs
Model Coeffs
Order (k)
4. Singularity Map
Ideal (b=0.500)
Target (b=-0.500)
Model (b=-0.023)
Poles ±0.5i
log10|Output|



实验结论

符合预期：目标伯恩斯坦椭圆区域越窄（beta越小），拟合效果越好